

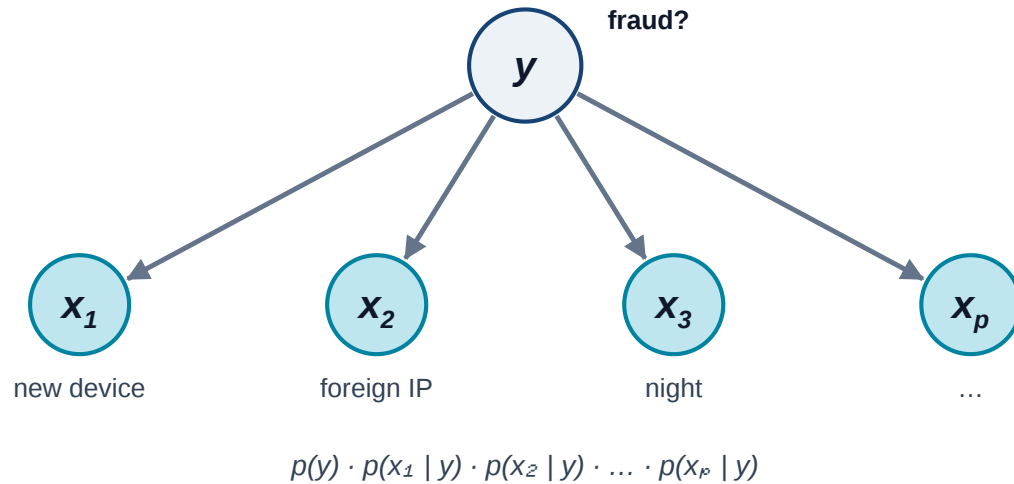
# Probabilistic Machine Learning

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## Lecture 05: Bayesian Networks

Prof. Dr. Michael Granitzer

# Where We Left Off



Lecture 04 made the joint affordable with one blanket assumption: **every** attribute independent of every other, given the class. It worked — and it counted correlated evidence twice.

What if we assumed only the independences we actually believe?

**Shortcut 4 — graph the joint.** Draw which variables depend on which; the graph *is* the factorisation. Dropped: every dependence **not drawn** — and, for a dense graph, affordable inference.

# Argument Map

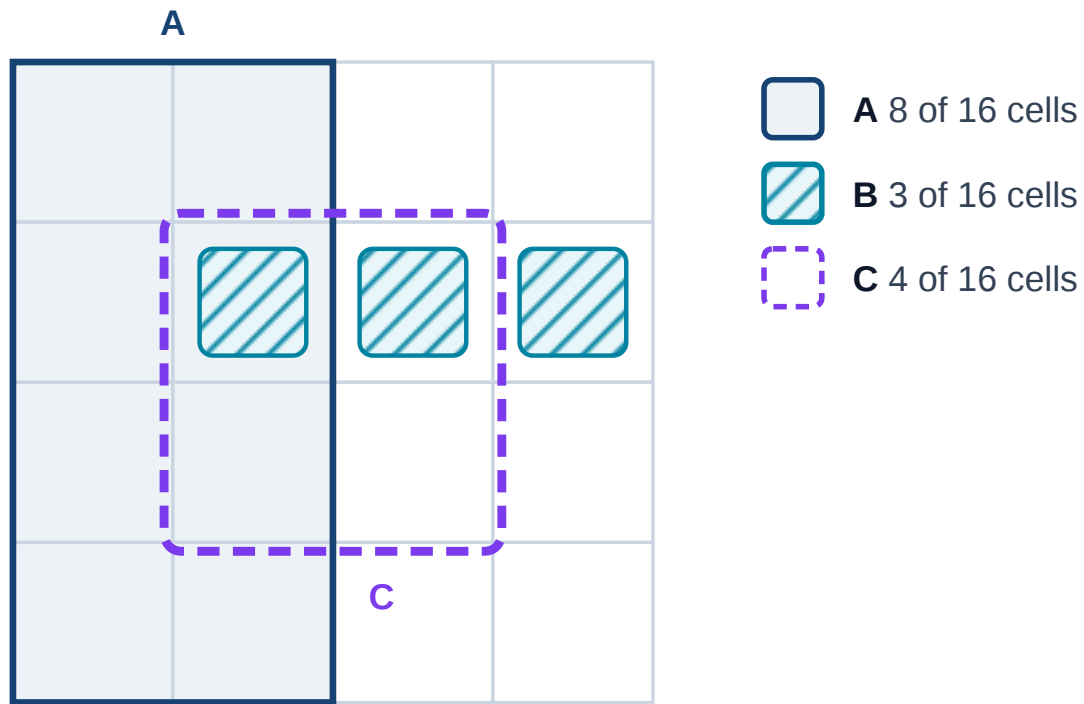


# Graph the Joint

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**Guiding Question:** Naive Bayes assumed every attribute independent of every other. What if we stated only the independences we believe — and what does conditioning then do to them?

# Independent Given C — Counted



Sixteen equally likely cells.

## Dependent

$$P(A \cap B) = \frac{1}{16} \neq \frac{3}{32} = P(A)P(B)$$

## Independent given $C$

$$P(A \cap B \mid C) = \frac{1}{4} = P(A \mid C)P(B \mid C)$$

# Why Is the Grass Wet?

Tracey sees her lawn is wet. **Rain — or did she leave the sprinkler on?** Then she sees that her neighbour Jack's lawn is wet too.

| variable | meaning (1 = yes)         |
|----------|---------------------------|
| $R$      | it rained                 |
| $S$      | Tracey's sprinkler was on |
| $J$      | Jack's grass is wet       |
| $T$      | Tracey's grass is wet     |

Before any arithmetic: does Jack's wet lawn make the sprinkler **more** or **less** likely?

# Background Knowledge as Independence

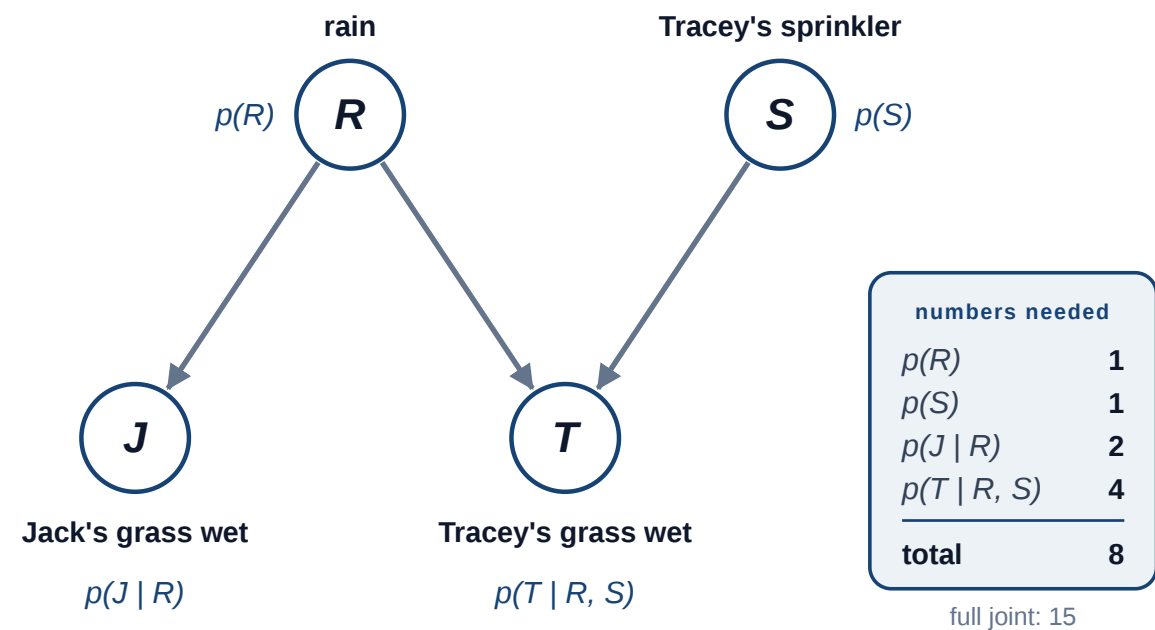
The joint needs  $2^4 - 1 = 15$  numbers. The chain rule rewrites it without shrinking it:

$$p(T, J, R, S) = p(T \mid J, R, S) \cdot p(J \mid R, S) \cdot p(R \mid S) \cdot p(S)$$

- Tracey's lawn ignores Jack's, given  $R$  and  $S$ :  $p(T \mid J, R, S) = p(T \mid R, S)$
- Jack's lawn ignores Tracey's sprinkler:  $p(J \mid R, S) = p(J \mid R)$
- the sprinkler does not change the weather:  $p(R \mid S) = p(R)$

$$p(T, J, R, S) = p(T \mid R, S) \cdot p(J \mid R) \cdot p(R) \cdot p(S)$$

# The Graph Is the Factorisation

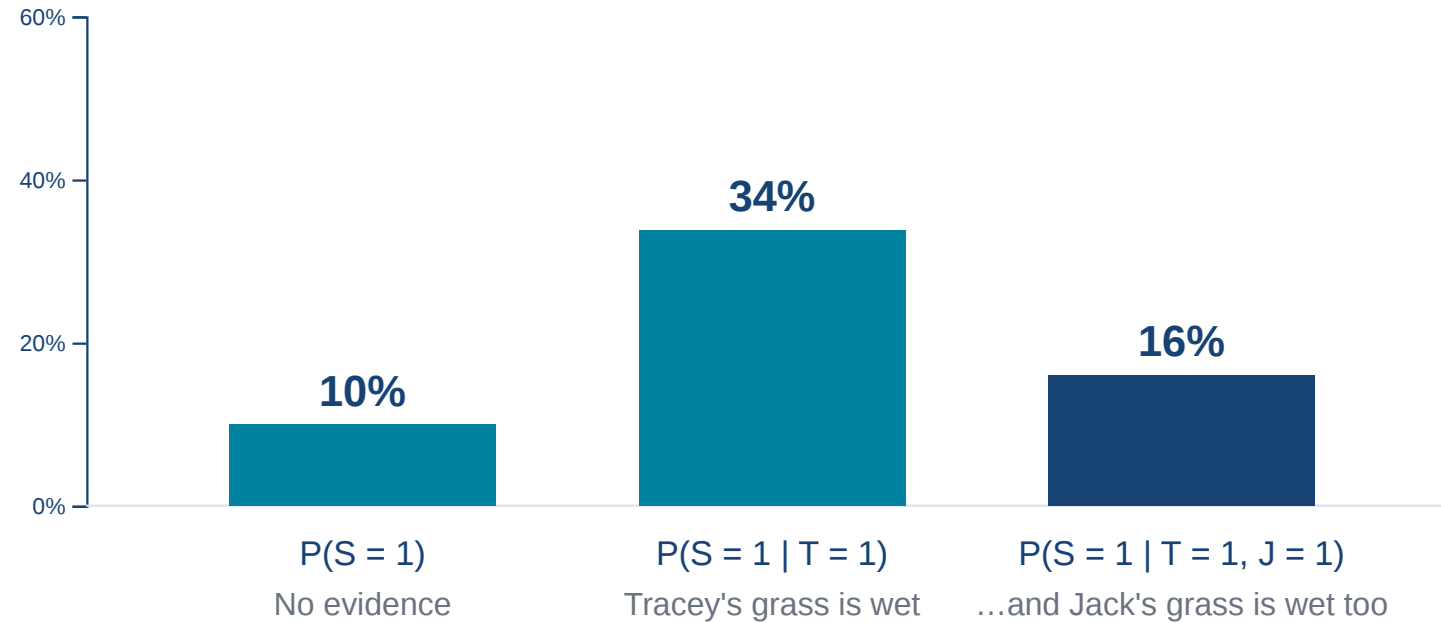


| factor                       | = 1 |
|------------------------------|-----|
| $p(R = 1)$                   | 0.2 |
| $p(S = 1)$                   | 0.1 |
| $p(J = 1 \mid R = 0)$        | 0.2 |
| $p(J = 1 \mid R = 1)$        | 1.0 |
| $p(T = 1 \mid R = 0, S = 0)$ | 0.0 |
| $p(T = 1 \mid R = 0, S = 1)$ | 0.9 |
| $p(T = 1 \mid R = 1, S = 0)$ | 1.0 |
| $p(T = 1 \mid R = 1, S = 1)$ | 1.0 |

One node per variable, one arrow per conditioning bar, one table per node.

# What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



# Explaining Away

$$p(S = 1 \mid T = 1, J = 1) = \frac{\sum_R p(T=1 \mid R, S=1)p(J=1 \mid R)p(R)p(S=1)}{\sum_{R,S} p(T=1 \mid R, S)p(J=1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Rain and sprinkler are independent:  $R \perp\!\!\!\perp S$ . Once their **common effect**  $T$  is observed they compete to explain it, and evidence for one lowers the other. Conditioning on a common cause removes a dependence; conditioning on a common effect **creates** one.

# Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

# Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

**Less.** The sprinkler explains the wet lawn.  $P(R = 1 \mid T = 1) \approx 0.74$  falls to  $P(R = 1 \mid T = 1, S = 1) = 0.02/0.092 \approx 0.22$  — almost back to the prior 0.2. Explaining away works in both directions.

# Takeaway

**Graph the joint:** background knowledge becomes conditional independence, and the graph is a picture of how the joint splits — 8 numbers instead of 15. Every *missing* edge is a claim; conditioning can make variables independent (common cause) or dependent (common effect).

# What an Arrow Claims

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**Guiding Question:** What exactly does an arrow in a probabilistic model claim — and what does a *missing* arrow claim?

# Belief Networks

**Bayesian belief network.** A **directed acyclic graph** whose nodes are random variables, with one conditional table per node given its parents:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i \mid \text{pa}(x_i))$$

**An arrow permits a dependence. A missing arrow claims an independence:** each variable is independent of its non-descendants, given its parents (the **local Markov property**).

Which independence does the wet-grass graph claim for Jack's lawn  $J$ ?

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Which independence does the wet-grass graph claim for Jack's lawn  $J$ ?

$J \perp\!\!\!\perp \{S, T\} \mid R$ : once we know whether it rained, nothing else says more about Jack's lawn.

# Counting the Numbers

| model, binary variables          | full joint                          | graph                        |
|----------------------------------|-------------------------------------|------------------------------|
| wet grass, 4 nodes               | 15                                  | $1 + 1 + 2 + 4 = 8$          |
| naive Bayes, class + 30 flags    | $2^{31} - 1 \approx 2.1 \cdot 10^9$ | $1 + 2 \cdot 30 = 61$        |
| 20 nodes, at most 3 parents each | $2^{20} - 1 \approx 10^6$           | at most $20 \cdot 2^3 = 160$ |

Where does the saving come from — and what makes it disappear?

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Where does the saving come from — and what makes it disappear?

A node with  $k$  binary parents needs  $2^k$  numbers. The saving comes from **missing arrows**; it disappears as soon as some node has many parents.

# Keeping Tables Small: Noisy-OR

A symptom  $y$  with many possible causes  $x_1, \dots, x_k$ : **each present cause fires on its own** with probability  $\lambda_i$ , and  $y$  occurs if any fires:

$$p(y = 0 \mid x_1, \dots, x_k) = \prod_{i: x_i=1} (1 - \lambda_i)$$

$k$  numbers instead of  $2^k$ . Two causes present,  $\lambda = 0.8$  and  $0.6$ :

$$p(y = 1) = 1 - 0.2 \cdot 0.4 = 0.92.$$

## THE ASSUMPTION BOUGHT

Causes act **independently** — no interaction. Two drugs that are harmless alone but dangerous together break it.

**Divorcing** is the other trick: insert a summary node ("total load") between a group of parents and the child.

# Mini-Exercise

Tracey and Jack share a garden hose: add the arrow  $S \rightarrow J$  to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

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How many numbers does the network need now — and which independence claim is gone?

**10.**  $p(J \mid R, S)$  needs 4 numbers instead of 2:  $1 + 1 + 4 + 4$ . Gone is  $J \perp\!\!\!\perp S \mid R$  — the missing arrow was that claim.

# Takeaway

A belief network is a **DAG plus one table per node**. An arrow only permits a dependence; every **missing** arrow claims an independence, and each claim saves numbers. Shrinking tables further — noisy-OR, divorcing — buys more savings with more assumptions, which should be said out loud.

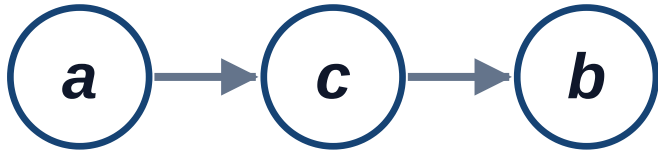
# Reading Independence off the Graph

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**Guiding Question:** Two causes are independent until you observe their shared effect.  
Can we see such things in the picture — before computing a single number?

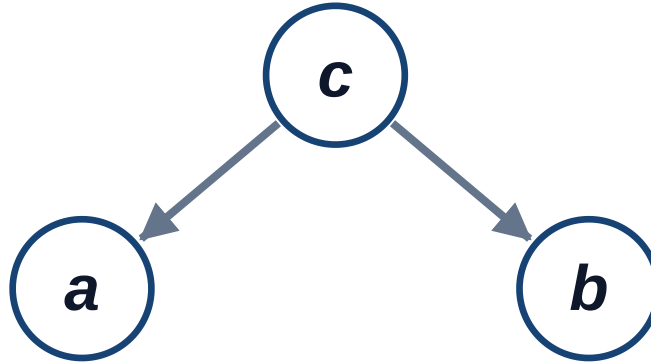
# Three Ways Three Nodes Connect

CHAIN



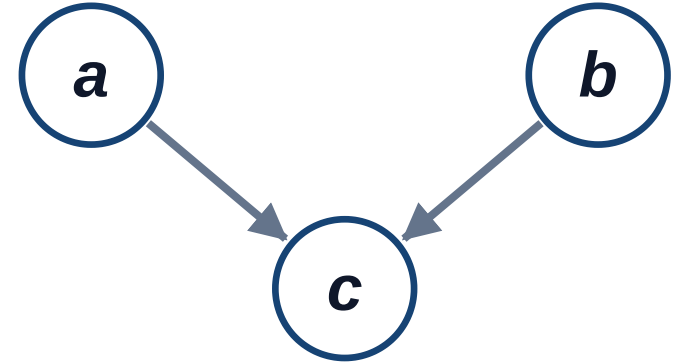
a, b: **dependent**  
given c: **independent**  
observing c blocks the path

FORK · COMMON CAUSE



a, b: **dependent**  
given c: **independent**  
observing c blocks the path

COLLIDER · COMMON EFFECT



a, b: **independent**  
given c: **dependent**  
observing c opens the path

Every independence question in a DAG reduces to these three.

# d-Separation

A path between  $X$  and  $Y$  is **blocked** by a set  $Z$  if it contains

- a chain or fork whose middle node **is in**  $Z$ , or
- a collider whose middle node — and every descendant of it — is **not in**  $Z$ .

If every path is blocked,  $X$  and  $Y$  are **d-separated** by  $Z$ , and then  $X \perp\!\!\!\perp Y \mid Z$  in every distribution that factorises over the graph.

Sound, not complete: a particular table may add independences the graph does not show — in our tables, once it rains, Tracey's lawn ignores the sprinkler.

# d-Separation on the Wet Grass

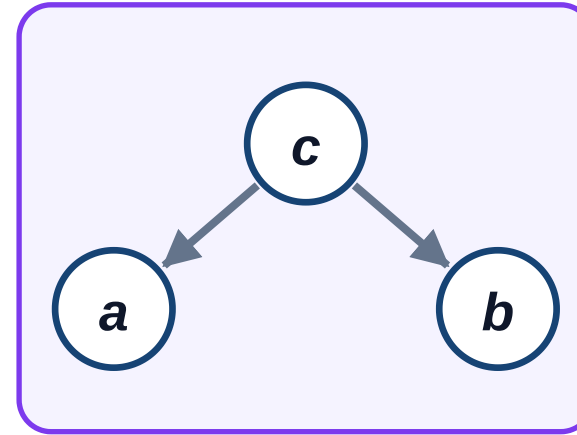
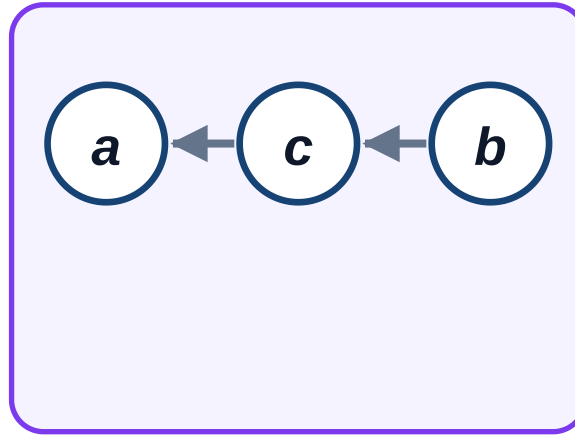
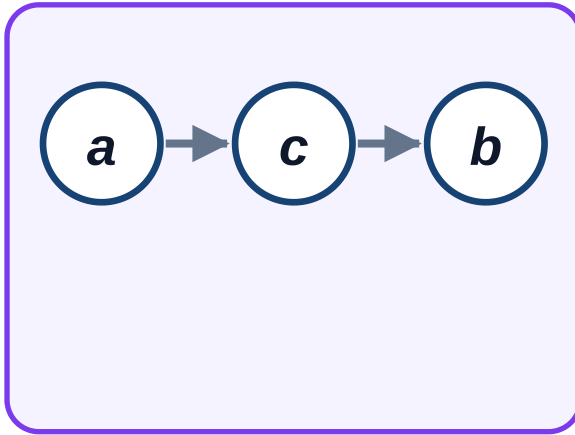
The only path from  $J$  to  $S$ :  $J \leftarrow R \rightarrow T \leftarrow S$  — a fork at  $R$ , a collider at  $T$ .

| query                              | path                             | verdict     | the numbers                                          |
|------------------------------------|----------------------------------|-------------|------------------------------------------------------|
| $J \perp\!\!\!\perp S ?$           | collider $T$ unobserved: blocked | independent | $P(J=1) = 0.36 = P(J=1 \mid S=1)$                    |
| $J \perp\!\!\!\perp S \mid T ?$    | collider $T$ observed: open      | dependent   | $P(J=1 \mid T=1) = 0.79$ , but 0.37 if also $S=1$    |
| $J \perp\!\!\!\perp S \mid T, R ?$ | fork $R$ observed: blocked       | independent | $P(S=1 \mid T=1, R=1) = 0.1$ , with or without $J=1$ |

The middle row is Module 1's explaining away: 10%  $\rightarrow$  34%  $\rightarrow$  16% for the sprinkler. The graph predicted it before any number was computed.

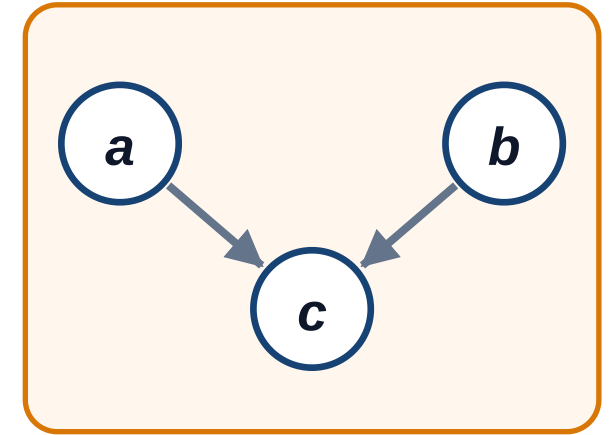
# Arrows Are Not Causes

same claim:  $a, b$  independent given  $c$



data cannot tell these three apart  
the direction of the arrows is not in the data

different claim



$a, b$  independent;  
dependent given  $c$

Same skeleton, same colliders: **the same independences** (Markov equivalence). Data cannot orient such arrows — they are **not** causal claims.

# Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add  $T \rightarrow D$ . You see only a **wet doormat** — not the lawn.

Are rain and sprinkler still independent?

# Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add  $T \rightarrow D$ . You see only a **wet doormat** — not the lawn.

Are rain and sprinkler still independent?

**No.**  $D$  is a descendant of the collider  $T$ , so observing it opens the path  $R \rightarrow T \leftarrow S$ : a wet doormat also makes rain and sprinkler compete to explain it.

# Takeaway

Independence in a belief network is a **property of the graph**: chains and forks are blocked by observing the middle, colliders are opened by it (or by a descendant). d-separation applies that to every path. What it cannot read off is the direction of cause: equivalent graphs make the same claims.

# Drawing Repeated Structure

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**Guiding Question:** A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn — and what does the drawing claim about the observations?

# Drawing a Model

A probabilistic model is a factorisation, and the diagram is its picture. Five conventions carry almost all of it:



**shaded**

observed: in the data



**open**

latent or parameter



**solid dot**

fixed, not inferred



**arrow**

child conditioned on parent



**plate**

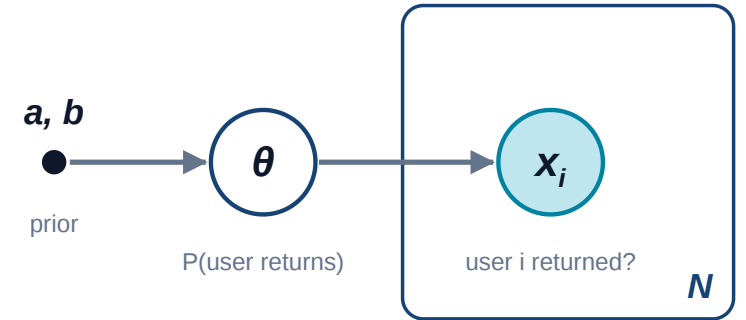
repeat N times

Nothing new is claimed: the arrows are read exactly as in a belief network. The shading only says **what the data fixes** — which variables we condition on.

# Plates

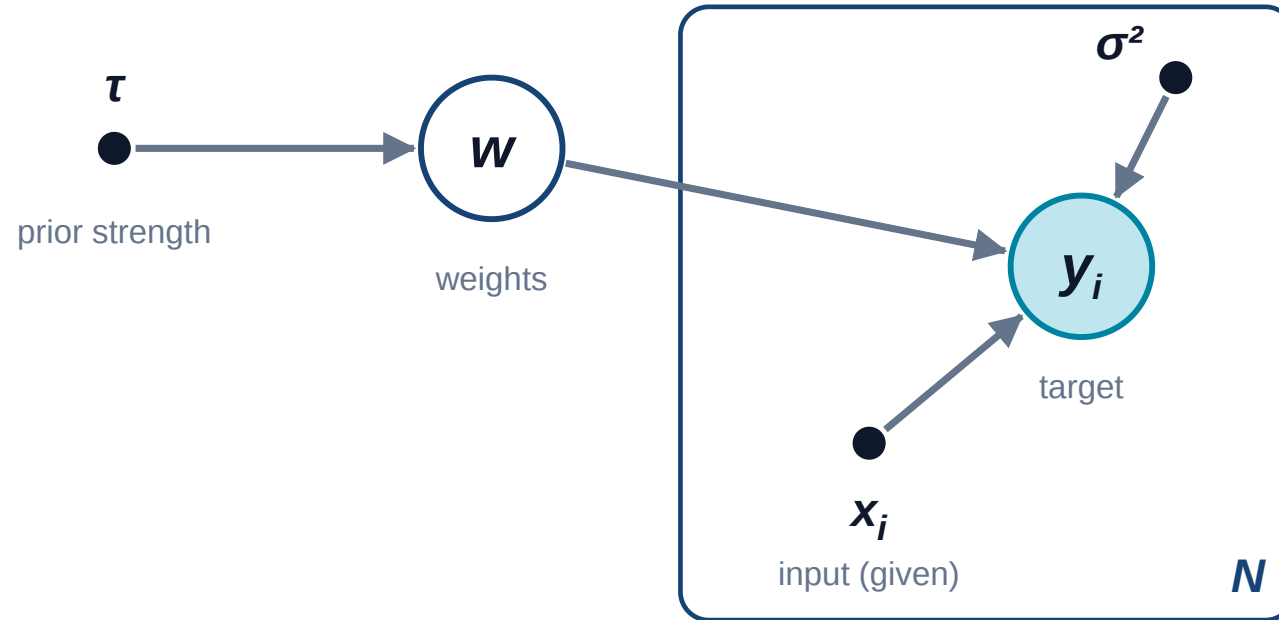
A **plate** is a box with a count in its corner: *repeat what is inside  $N$  times*. A model of 10,000 users does not need 10,000 nodes.

The copies are **independent given what sits outside** — the parameters, which learning estimates. Nested plates draw grouped data:  $K$  hospitals with  $N_k$  patients each.



Given  $\theta$  the users are independent. With  $\theta$  unknown they are **not**: in Lecture 02, the first ten users changed our prediction for the eleventh. That dependence is exactly why learning from data works.

# Reading a Diagram



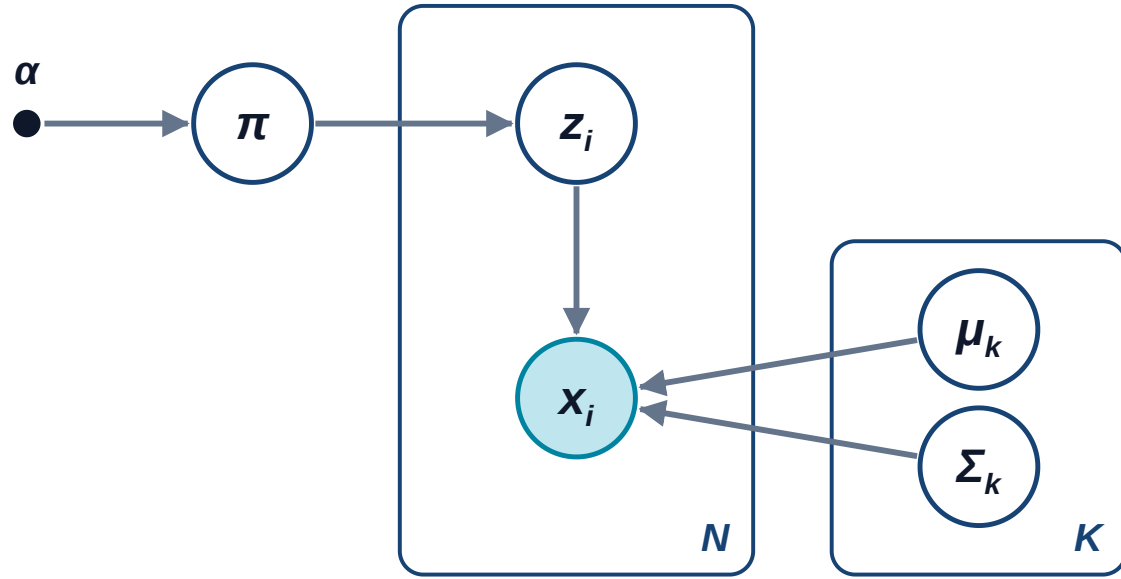
Read off the factorisation:

$$p(w \mid \tau) \prod_{i=1}^N p(y_i \mid x_i, w, \sigma^2)$$

The plate became the product; each arrow became a conditioning bar.

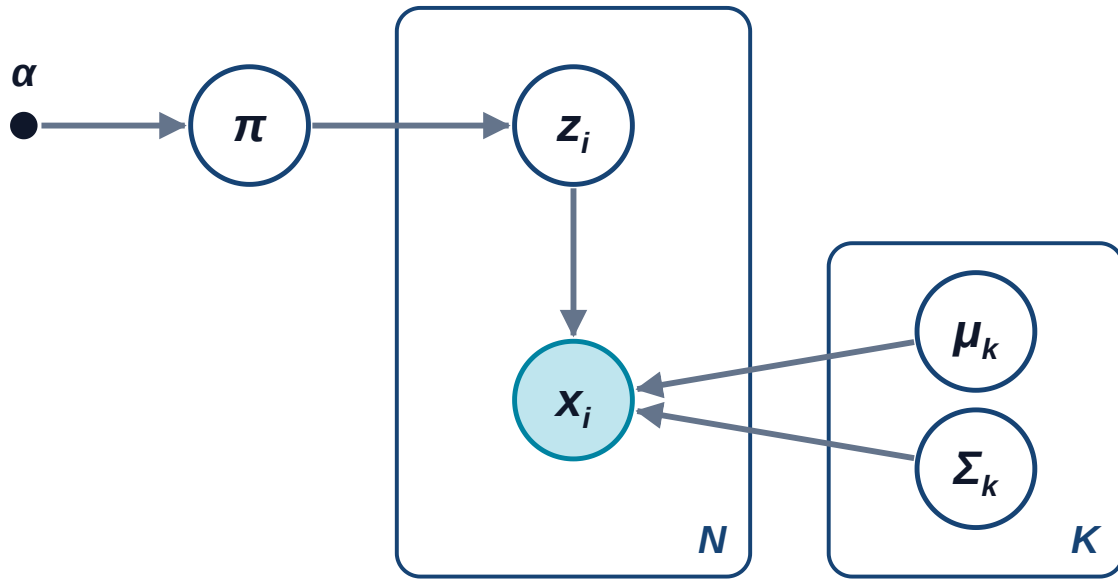
$x_i$  is a **dot**: the inputs are given, not modelled — this is a discriminative model,  $p(y \mid x)$ , with a prior on  $w$ . Lecture 07 conditions on the shaded  $y_i$  and keeps the whole posterior over  $w$ .

# Mini-Exercise



Which nodes does learning have to infer — and why is  $\alpha$  not among them?

# Mini-Exercise



Which nodes does learning have to infer — and why is  $\alpha$  not among them?

$\pi$ , every  $\mu_k, \Sigma_k$  and every label  $z_i$  — all **open**.  $\alpha$  is a **dot**: fixed by us. Only the  $x_i$  are observed:

$$p(\pi \mid \alpha) \prod_k p(\mu_k, \Sigma_k) \prod_i p(z_i \mid \pi) p(x_i \mid z_i, \mu, \Sigma)$$

**A mixture of Gaussians** — Lecture 07.

# Takeaway

A model diagram is a factorisation you can see: **shading** marks what the data fixes, **plates** mark what repeats independently *given* what sits outside, and **what sits outside** is what learning has to estimate.

# Can We Compute With It?

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**Guiding Question:** We can now write the joint down. Can we actually condition on it — and where do its numbers come from?

# Three Questions You Can Ask a Model

| question                          | on the wet grass     | with complete data                   |
|-----------------------------------|----------------------|--------------------------------------|
| <b>infer</b> unobserved variables | $P(S \mid T, J)$     | cost set by the graph's shape        |
| <b>learn the parameters</b>       | the 8 table entries  | counting                             |
| <b>learn the structure</b>        | which arrows at all? | hardest — and only up to equivalence |

# Inference by Elimination

Explaining away needed  $P(T=1, J=1)$ . Sum the factorised joint — and **push each sum as far right as it goes**:

$$P(T=1, J=1) = \sum_R p(R)p(J=1 \mid R) \underbrace{\sum_S p(S)p(T=1 \mid R, S)}_{m(R)}$$

## Inner sum first

$$m(1) = 0.1 \cdot 1 + 0.9 \cdot 1 = 1$$

$$m(0) = 0.1 \cdot 0.9 + 0.9 \cdot 0 = 0.09$$

## Then the outer one

$$0.2 \cdot 1 \cdot 1 + 0.8 \cdot 0.2 \cdot 0.09$$

$$= 0.2 + 0.0144 = 0.2144 \quad \checkmark$$

# Why Pushing Sums In Pays

A chain of  $n$  binary variables  $X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_n$ . To get  $p(x_n)$ :

$$2^{n-1}$$

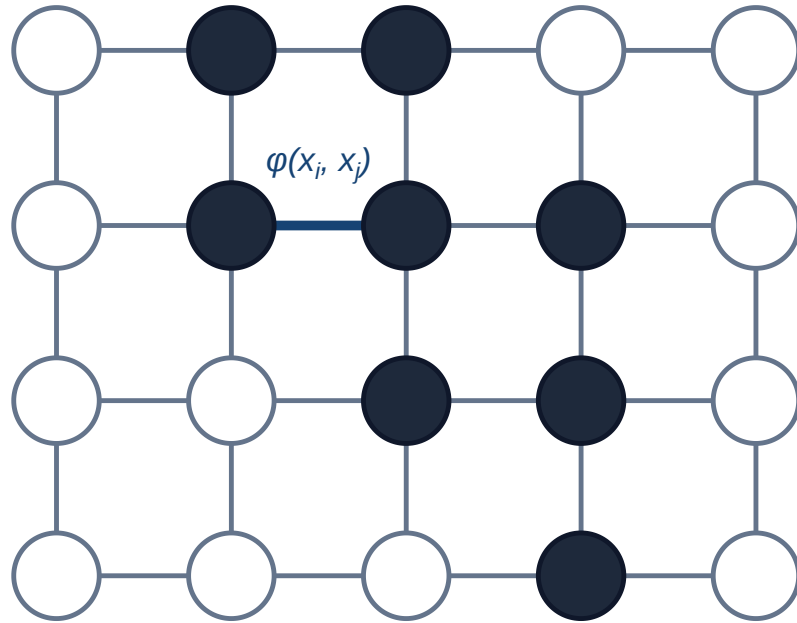
terms if every other variable is summed at once — for  $n = 30$ , about 537 million

$$4(n - 1)$$

operations if the sums are pushed in, one variable at a time — for  $n = 30$ , 116

Eliminating  $X_1$ , then  $X_2$ , ... passes a **two-entry table** along the chain. The cost is set by the **largest table** created on the way — by the graph's shape, not its size.

# Undirected Models



each edge: a potential that prefers neighbours to agree

Pixels influence each other **symmetrically**. A **Markov network** puts a non-negative **potential**  $\phi$  on each edge and normalises once, globally:

$$p(x) = \frac{1}{Z} \prod_{(i,j)} \phi(x_i, x_j), \quad Z = \sum_x \prod_{(i,j)} \phi(x_i, x_j)$$

Two pixels,  $\phi = 3$  if equal, 1 if not:  $Z = 3 + 1 + 1 + 3 = 8$ , so  $P(\text{equal}) = 6/8$ .

# Directed and Undirected

|                           | belief network                         | Markov network                            |
|---------------------------|----------------------------------------|-------------------------------------------|
| edges                     | directed, acyclic                      | undirected                                |
| factors                   | conditional tables, normalised locally | potentials, one global $Z$                |
| natural for               | causal or generative stories           | symmetric interaction: pixels, neighbours |
| observing a common effect | creates dependence                     | — (no colliders)                          |

Inference runs on the undirected form: **moralisation** joins parents that share a child ( $R - S$  in the wet grass) and drops the arrow heads; elimination works on that graph.

# Writable Is Not Queryable

A  $100 \times 100$  black-and-white image as a pixel lattice:

**79,200**

numbers to **write it down**: 19,800 edges, 4 per potential

$2^{100} \approx 10^{30}$

entries in the table that elimination must build — a whole row of pixels at once

Exact inference in belief networks is **NP-hard** in general, and so is approximate inference with a guaranteed error bound. What remains: exploit the shape where it is kind (chains, trees), and otherwise **sample** or **approximate** — trading the guarantee for an answer.

# Learning the Parameters Is Counting

Fifty observed days. Estimate Jack's table  $p(J=1 \mid R)$ :

|         | days | Jack's lawn wet | MLE (peak)    | +1 pseudo-counts (mean) |
|---------|------|-----------------|---------------|-------------------------|
| $R = 0$ | 40   | 8               | $8/40 = 0.20$ | $9/42 \approx 0.21$     |
| $R = 1$ | 10   | 10              | $10/10 = 1.0$ | $11/12 \approx 0.92$    |

Counting **is** maximum likelihood (L03); pseudo-counts **are** the Beta posterior mean (L02) — Laplace smoothing (L04). "Rain always wets the grass" was an MLE on ten days.

**Hidden** variables leave no counts: infer them, re-count, repeat — EM, Lecture 07.

# Learning the Structure

## Score and search

Try graphs, score each by how well it fits the data *per parameter*, keep the best. The number of DAGs grows super-exponentially.

## Test independences

Run conditional-independence tests and keep the graphs consistent with them — d-separation, read backwards.

Either way, data determines a graph only **up to Markov equivalence**: arrows between equivalent graphs are not in the data. A learned structure is not a causal discovery.

# Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for  $p(J=1 \mid R=1)$  — and what does one pseudo-count on each side give?

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Jack's lawn was wet on all **4** rainy days in a small sample.

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**Counting:**  $4/4 = 1$  — "rain always wets the lawn", from four days. **Smoothed:**  $5/6 \approx 0.83$ , the posterior mean under  $\text{Beta}(1, 1)$ . Same data, two readings: the peak and the mean.

# Takeaway

A graphical model makes the joint **writable**; its **shape** decides whether it is **queryable** — elimination is cheap on chains and trees and hopeless on a lattice. With complete data the parameters are counts: the MLE, or with pseudo-counts the Beta posterior mean. The structure is learnable only up to equivalence.

# Appendix

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# Uncertain and Unreliable Evidence

## Uncertain (soft) evidence

We are not sure of the observation itself: a distribution  $\tilde{x}$  over its states. Sum over them:

$$p(y \mid \tilde{x}) = \sum_x p(y \mid x)p(x \mid \tilde{x})$$

*We look at the lawn from the window without touching it:  $p(T=1 \mid \tilde{T}) = 0.8$ .*

## Unreliable evidence

We keep the report but **replace** its table by our own judgement of how reliable it is:

$$p(x \mid \text{pa}(x)) \rightarrow \tilde{p}(x \mid \text{pa}(x))$$

*Jack says his lawn is wet, but he did not check.*

Both are drawn with a **dashed** edge.